Data Structure

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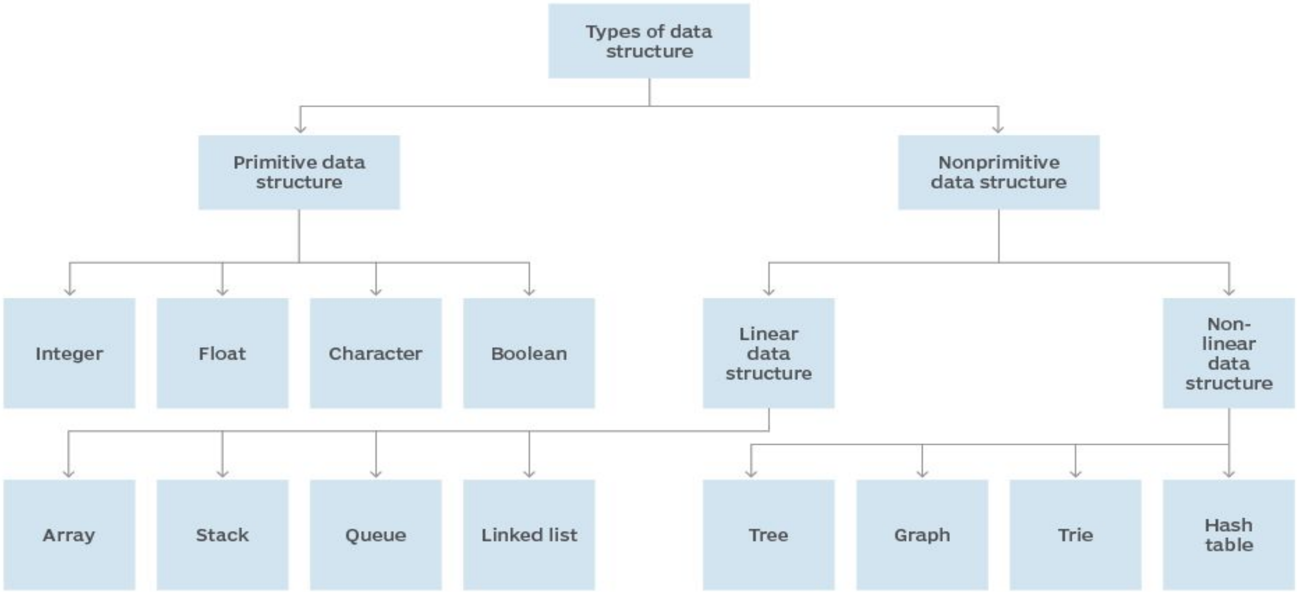
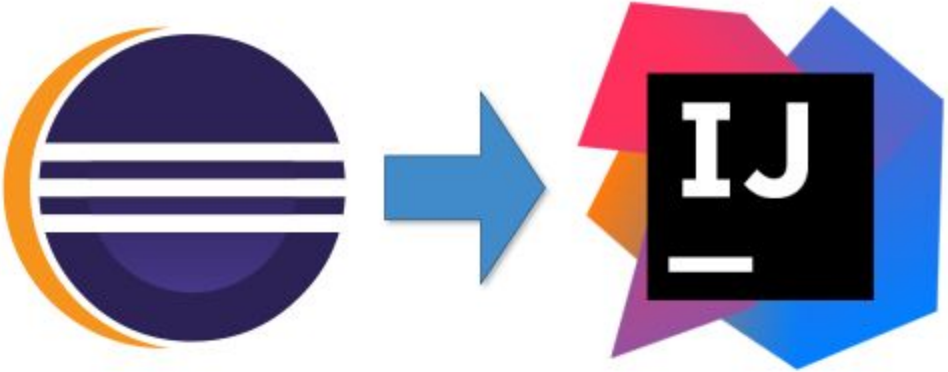
Last class.....

Data structure hierarchy



Student must need to use IDE

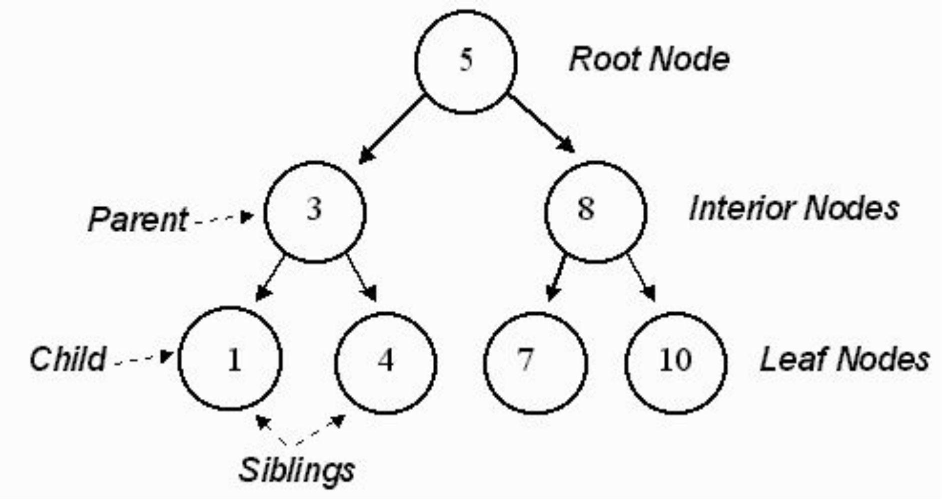


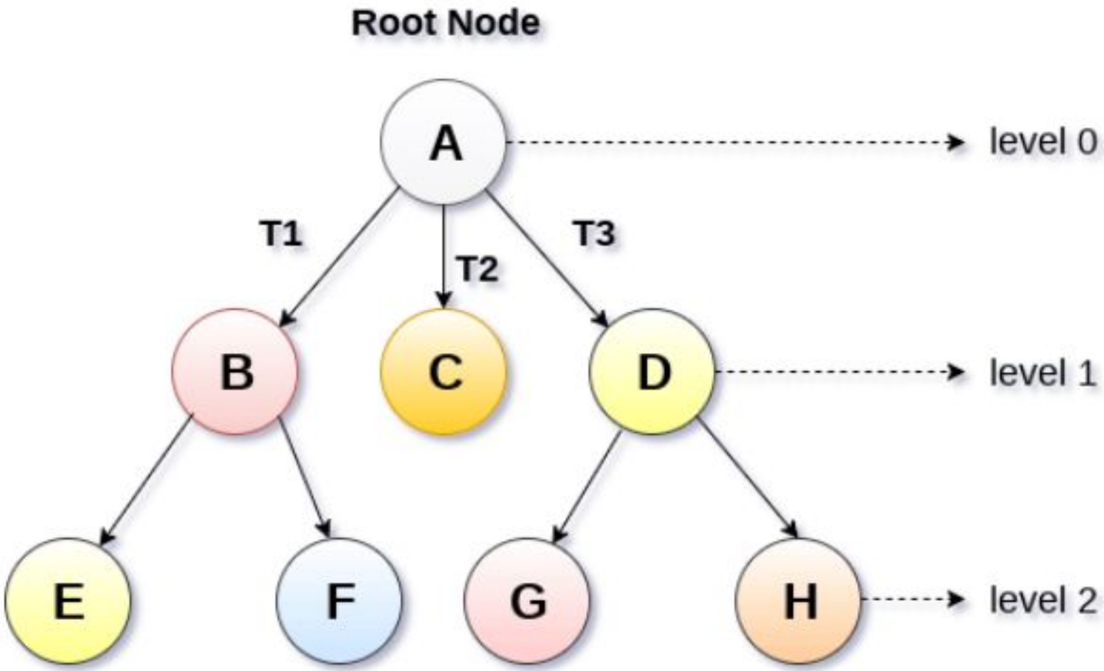
Light weight Online Java Compiler - https://www.codiva.io/
https://paiza.io/en/projects/new?language=java

Today's Topic: Trees

Tree

We extend the concept of linked data structures to structure containing nodes with more than one self-referenced field. A tree is made of nodes, where each node contains a "left" reference, a "right" reference, and a data element. The topmost node in the tree is called the root.




Root Node
A
T1
T2
T3
B
C
D
E
F
G
H
level 0
level 1
level 2

Static representation of tree

```
#define MAXNODE 500
struct treenode {
    int root;
    int father;
    int son;
    int next;
}
```

Dynamic representation of tree

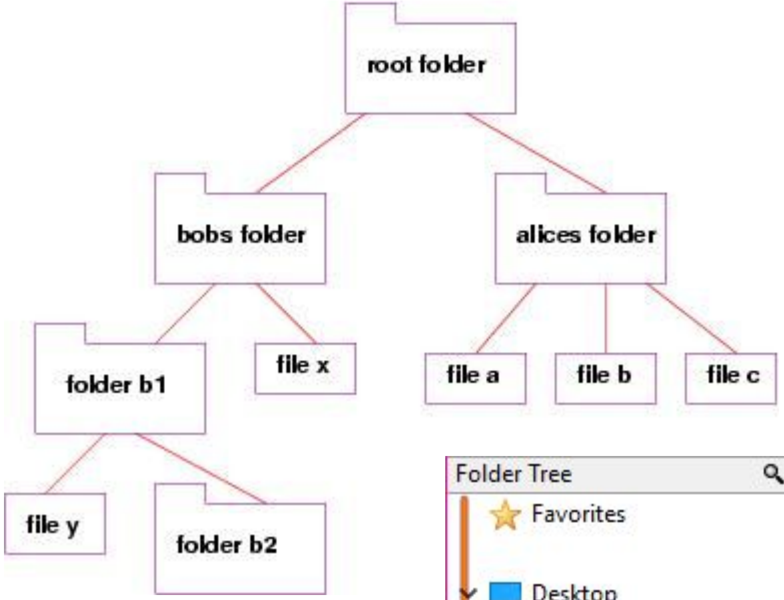
```
struct treenode
{
    int root;
    struct treenode *father;
    struct treenode *son
    struct treenode *next;
}
```

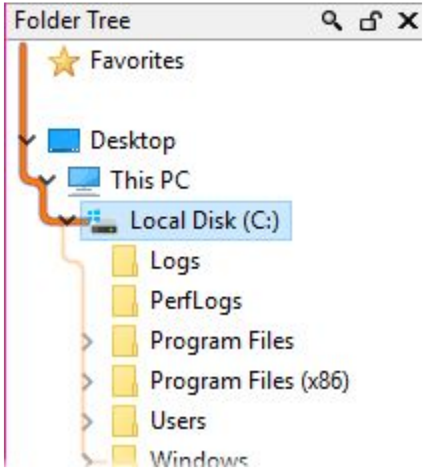
Advantages of trees

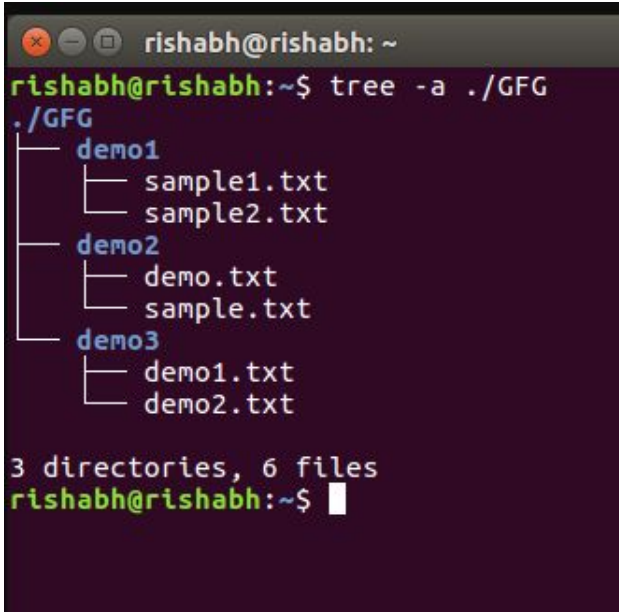
Trees are so useful and frequently used, because they have some very serious advantages:

- Trees reflect structural relationships in the data
- Trees are used to represent hierarchies
- Trees provide an efficient insertion and searching
- Trees are very flexible data, allowing to move subtrees around with minumum effort

Examples of trees







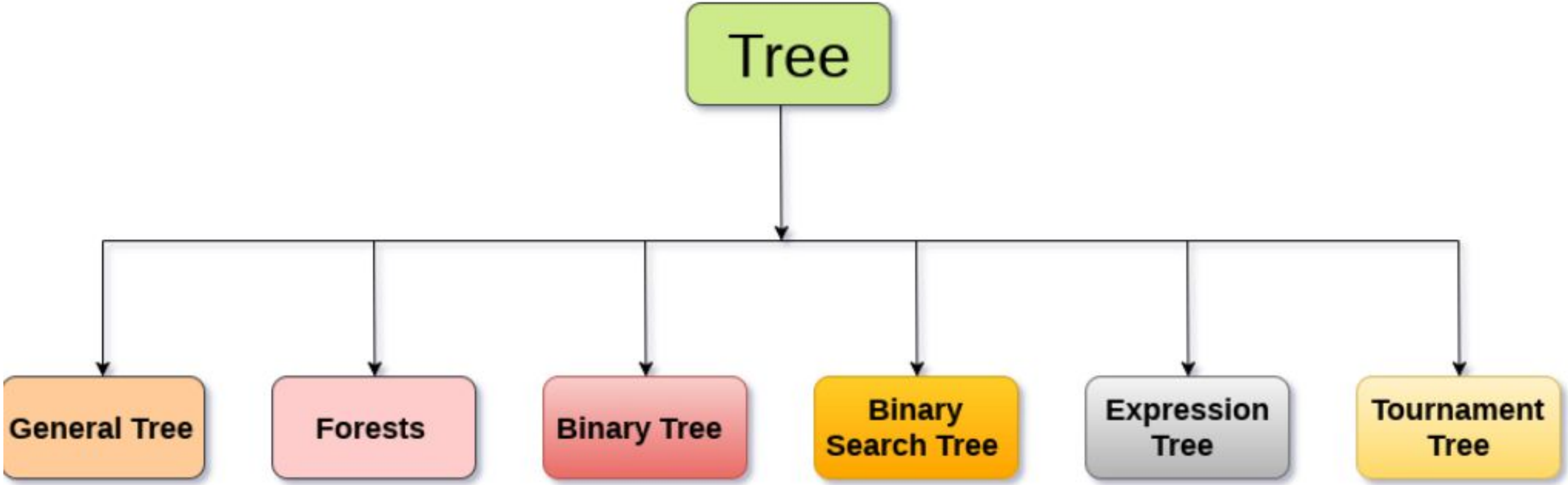
Implementation Tree

```c
struct node {
   int data;
   struct node *leftChild;
   struct node *rightChild;
};
```

```java
class Node {
    int value;
    Node left;
    Node right;

    Node(int value) {
        this.value = value;
        right = null;
        left = null;
    }
}
```

Types Of Tree

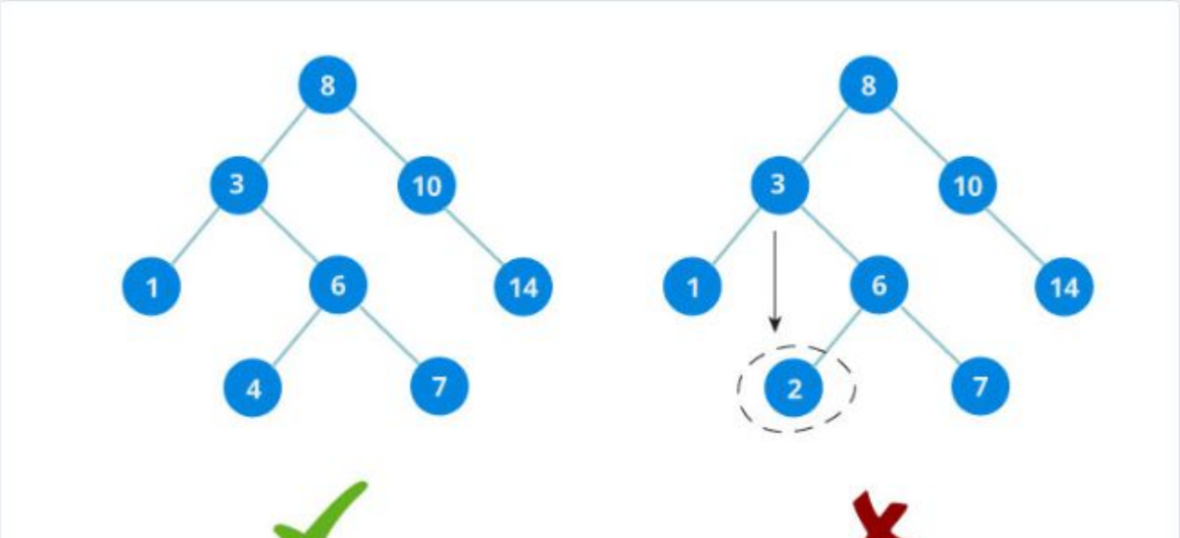


Binary Search

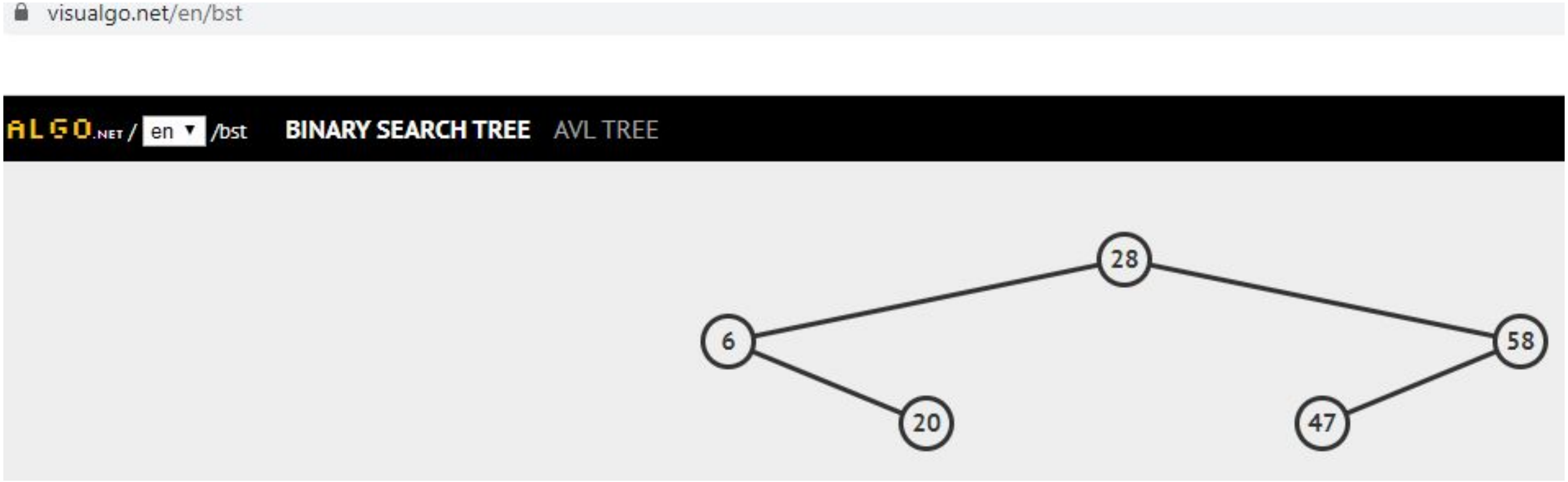
- It is called a binary tree because each tree node has maximum of two children.
- It is called a search tree because it can be used to search for the presence of a number in `O(log(n))` time.

The properties that separates a binary search tree from a regular binary tree is

1. All nodes of left subtree are less than root node
2. All nodes of right subtree are more than root node
3. Both subtrees of each node are also BSTs i.e. they have the above two properties



Binary Search Tree



https://visualgo.net/en/bst

Algorithm

```
If root.data is equal to search.data
   return root
else
   while data not found

      If data is greater than node.data
         goto right subtree
      else
         goto left subtree

      If data found
         return node
   endwhile

   return data not found

end if
```

```c
struct node* search(int data) {
   struct node *current = root;
   printf("Visiting elements: ");

   while(current->data != data) {
      if(current != NULL)
      printf("%d ",current->data);

      //go to left tree

      if(current->data > data) {
         current = current->leftChild;
      }
      //else go to right tree
      else {
         current = current->rightChild;
      }

      //not found
      if(current == NULL) {
         return NULL;
      }

      return current;
   }
}
```

Home Task: Write 2 binary search example program

Examples here - https://www.geeksforgeeks.org/binary-tree-set-1-introduction/